EFY TESTED
NIDHI KATHURIA & NAMAN GARG

Simple 8-BIT COMPUTER For Learning

VINEET NAYAK

Computers take an input, process it as per a set of instructions and provide the output. These are general-purpose devices that can be programmed to carry out a set of arithmetic or logical operations automatically.

Intel 8008 was the first 8-bit microprocessor for the microcomputer system. The architecture of a microprocessor seems complex, but a computer can be entirely built with TTL logic chips without using any microprocessor/controller. This project helps you build your own computer while also providing an insight into the working of a general-purpose microprocessor as well as its construction mechanism.

Architecture

This 8-bit digital TTL computer is designed as per the architecture shown in Fig. 1. The entire processor circuit is powered from an SMPS as it consumes a lot of power. Voltage supply is 5V.

The architecture has a common 8-bit bidirectional data/address bus. The arithmetic logic unit (ALU) consists of adder/subtractor circuit along with AND, OR, NOT and XOR gates to perform logical and arithmetical operations. The values for operation are stored in A-register (Accumulator) and B-register. The number of states and sequence of states for all the instructions are stored in the ROM. Two ROMs are used to control the sequence of operation for various modules in the system. ROMs provide 16 control lines for operation.

The output and enabling of different modules are controlled by EN pins of ICs or tristate buffers. For example, the ALU output is controlled by using tristate buffer at the end of each logic output. The ALU output selection based on the instruction is done by a decoder connected to ROM2 as shown in the circuit diagram (Fig. 2 may not be clearly visible here because of its large size so it has been included in the EFY DVD and can also be downloaded from *source.efymag.com*).

Typically, the sequence of operation is as follows:

1. The program counter (PC) is incremented
2. The PC value is loaded in the memory address register
3. The instruction is stored in the instruction register of the RAM at the address specified by the memory address register
4. The PC is incremented
5. The memory address register is updated with a new PC value
6. The LDI instruction loads the data in the accumulator specified by the memory address of the RAM
7. The instruction register is cleared for next instruction loading

The author's prototype with five LED bar displays is shown in Fig. 3. LED bar displays show the status of output, data, address, instruction set and state. This helps in tracking how each instruction is being executed.

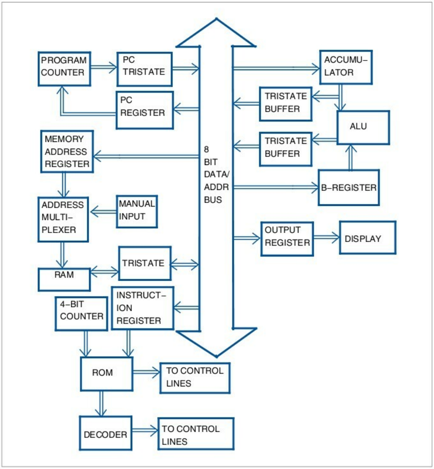

Fig. 1: Processor architecture